

Inventory & Library Management Module

Pre-School / Daycare Management System ke liye taiyar kiya gaya. Books, toys, stationery, uniform aur har physical item ke liye ek hi transparent system — library alag module nahi, isi ka hi ek hissa hai.

"Har item ko track karne ke liye ek simple, transparent system — aane se lekar use hone, issue hone, udhaar jaane ya bikne tak."

PROJECT

Brain Builder International – Backend API

MODULE

Inventory + Library (unified)

VERSION

4.0 – Final

STATUS

Draft for review

Contents

01	Introduction	06	Ek din ka example
02	Objective	07	Roles & access
03	Item categories	08	Key benefits
04	Key features (4.1–4.13)	09	Conclusion
05	Enquiry → Student		

01 Introduction

Har din ek pre-school / daycare mein bahut saari physical items use hoti hain — toys, books, stationery, craft material, uniform, school bag, gift items, aur bhi bahut kuch. Time ke saath naye-naye tarah ke items add hote rehte hain, isliye pehle se har cheez ki list bana pana possible nahi hai.

Isi problem ko solve karne ke liye ek hi module banaya ja raha hai — **Inventory & Library Management** — jahan har item, chahe kitna bhi chota ya bada ho, uske aane se lekar use hone, issue hone, udhaar jaane, bikne ya wapas aane tak har stage par record hoga.

Library alag system nahi hai — books bhi inventory ki hi ek item category hain, jinke liye lending (borrow/return) ka wohi flow use hota hai jo toys jaisi doosri lendable items ke liye ban raha hai. Ek hi document, ek hi system, ek hi Admin panel — aur poora scope ek hi baar mein taiyar hota hai, kisi phase-wise split ke bina.

02 Objective

- Har item ko record karega jo school mein aati hai — quantity, date aur source ke saath.
- Har item ko track karega jo store se bahar jaati hai — kisko mili, kitni, aur kab.
- Fixed time ke liye di gayi books/toys ke liye automatic reminder — yehi library ka core kaam hai.
- Admin ko pehle se low-stock alert dega.
- Teacher/parent ko apna khud ka item history dekhne dega — Admin se poochhe bina.
- Bulk Excel template se ek saath bahut saari items add karne dega.
- Kuch items ko public page par dikhayega, aur parents ko unhe **wishlist** karne dega.
- Kisi bhi item ya poori category ko class-wise **free ya chargeable** set karne dega.
- Ye ensure karega ki koi item **bikne ke liye tabhi available ho jab Admin explicitly ON kare** — by-default kuch bhi sale ke liye nahi khula rehta.
- Rented items ke liye ek **safety check** rakhega, taaki koi item uski value se kam deposit par galti se issue na ho.
- **Bahar ke logon ko bhi seedha Student ke roop mein convert karke access dega** — bina koi alag public signup banaye, existing Enquiry process reuse karke (Section 5).
- Admin ko Library aur Inventory, dono ke liye **alag-alag, category-wise ya item-wise permissions** delegate karne dega.

03 Kis tarah ki items cover hongii

CATEGORY	MATLAB	EXAMPLE
Consumable	Ek baar use hokar khatam, wapas nahi aati	Pencil, chart paper, glue, marker
Returnable / Lendable	Kuch time ke liye di jaati hai, wapas aane ki expectation	Books , toys, sports equipment
Sellable	Parents ke dwara khareedi jaati hai, sirf Admin ke "Available for Sale" ON karne par	Uniform, bag, T-shirt, gift items, permanent books

04 Key features

4.1 — Item Master

Har item ka apna record: naam, category, quantity unit, current stock, minimum stock level, storage location, photo, vendor details, selling/replacement price, aur (lendable items ke liye) security deposit amount. Available-for-Sale aur class-wise free settings bhi isi record ka hissa hain.

Books quantity ke hisaab se track hoti hain. Har copy ko alag accession number se track karna ek **optional, per-item setting** hai.

4.2 — Stock In

Naya stock aane par: kitne packet/box, har packet mein kitne piece, date, vendor, bill/invoice — save hote hi stock apne aap update.

4.3 — Stock Out / Issue

By-default har item ke liye Admin ki approval required hoti hai — teacher request bhejti hai, Admin approve karta hai, tabhi issue hoti hai. Admin chahe to kisi item ko is requirement se exempt kar sakta hai (direct-issue), do level par:

- **Poori category ke liye** — jaise saari "Books" ko direct-issue enable karna.
- **Single item ke liye** — jaise sirf "Pencil" aur "Eraser" ko individually enable karna.

Example: Pencil/Eraser direct-issue enabled hain, teacher seedha issue kar sakti hai. "Sports Equipment" category approval maangti hai — request Admin ke paas jaati hai.

4.4 — Parents ko items issue karna: Borrow ya Buy **LIBRARY KA CORE**

Fixed time ke liye borrow — issue date + return-by date record; time par wapas na aane par apne aap reminder. **Hamesha ke liye buy** — agar item Available for Sale hai, to sale ki tarah record, fee/payment side se connected.

Deposit safety check: agar item ki selling/replacement price uske security deposit se zyada hai, system issue nahi hone dega — issuing teacher/Admin ko turant alert milega.

4.5 — Available for Sale

Har item ke liye ek "Available for Sale" switch — **default hamesha "Not for Sale"**. Jab tak Admin explicitly ON na kare, koi item purchase ke liye offer nahi hoti.

4.6 — Class-wise Free / Chargeable Access

Har item — ya poori category — ke liye Admin decide karta hai ki wo kis class ke liye free hai, kis ke liye paid.

Example: "Books" category ko hi select karke Nursery/LKG ke liye free kiya ja sakta hai — har book ke liye alag se setting nahi karni padti. Ek specific item par apni khud ki setting bhi di ja sakti hai, jo category-level ko override karti hai.

4.7 — Public Page Visibility

Admin control karta hai ki item public page (website/home) par dikhega ya nahi, aur kya-kya info dikhegi. Internal detail (stock qty, vendor, cost) kabhi public nahi. Interested visitor seedha Enquiry submit kar sakta hai (Section 5).

4.8 — Wishlist

Parents kisi bhi item ko wishlist mein save kar sakte hain. Item stock mein wapas aate hi apne aap notification chala jaata hai.

4.9 — "My Items" personal panel

Jisko bhi item di jaati hai, wo apne aap unke apne panel par dikh jaati hai — kya diya, kitna, kab — saath mein wishlist bhi ek tab mein.

4.10 — Bulk Upload

Ready Excel template download → offline bharo → upload → system read karke fields bhar deta hai → final save se pehle screen par review → upload ke baad bhi har item editable.

4.11 — Role & Permission Based Access **LIBRARY + INVENTORY ALAG-ALAG**

Naya "librarian" role banane ke bajaye, existing teacher accounts par hi **category-wise granular** permissions di jaati hain.

Library permissions

- Catalog — items add/edit, pricing, visibility, available-for-sale
- Issue & Return — borrow/buy/return
- Reports — library reports hi

Inventory permissions

- Stock In — naya stock record
- Stock Out — jahan direct-issue enabled hai
- Catalog — item master add/edit
- Reports — inventory reports hi

Example: Ek teacher ko sirf "Library — Issue & Return" milta hai; ek doosri ko sirf "Inventory — Stock In." Admin dono groups sabko ek saath bhi de sakta hai.

4.12 — Notifications & Alerts

Low-stock, overdue return, upcoming due reminder, wishlist restock, issue-approval pending, aur deposit-mismatch alerts.

4.13 — Reports & Dashboard

Total stock value, category-wise breakdown, issued/borrowed + overdue list, sabse zyada use/borrow aur wishlist hone wali items, month ka expense.

05 Bahar ke log kaise judte hain — Enquiry se, seedha Student ke roop mein

Pehle library-only interest walon ke liye ek alag "**Library Member**" type ki soch thi. **Ye hataya ja raha hai** — system mein ab sirf ek hi conversion hai: **Student**.

Chahe enquiry poori tarah admission ke liye ho ya sirf library interest se, dono **wahi existing Enquiry process reuse karte hain** (**New** → **Contacted** → **Visited** → **Admitted / Rejected**), aur Admin usi "Convert to Student" action se enquiry ko seedha Student mein convert karta hai.

- 1 **Enquiry aati hai** — Admin se contact ya public page se seedha submit.
- 2 **Follow-up** — existing enquiry stages ke through.
- 3 **Convert to Student** — chahe full admission ho ya sirf library interest.
- 4 **Access milta hai** — student ke Parent ko turant My Items + wishlist access mil jaata hai.

Fayda: Ek hi pipeline, ek hi entity type — na alag Member-record, na do tarah ke access-level. Admin ke paas poora control, koi random khud se signup nahi kar sakta.

06 Ye kaise kaam karega — ek din ka example

- Story books ka carton aata hai → Stock In record → wishlist kiye hue parents ko "ab available" alert.
- Teacher "Pencil" (direct-issue enabled) seedha issue karti hai; wahi teacher "Sports Kit" (approval-required) ke liye request bhejti hai, Admin approve karta hai.
- Naya interested parent public page par toy dekh kar enquiry karta hai → Admin seedha Student mein convert kar deta hai.
- Wahi parent "Craft Kit" khareedte hain, jo poori "Books/Kits" category par Nursery ke liye free set hai → price zero dikhta hai.
- Doosra parent ek mehenga science kit borrow karna chahta hai jiski price deposit se zyada hai → system issue block karta hai, teacher ko alert milta hai.
- Ek book overdue ho jaati hai → apne aap reminder, Admin ke liye flag.
- Marker stock minimum se neeche → low-stock alert.
- Month-end → dashboard par expense, kam stock, sabse zyada borrow/wishlist hui items, outstanding — sab ek screen par.

07 Roles & access

ROLE	KYA KAR SAKTE HAIN	EXAMPLE
Admin	Full control — items, stock, pricing, available-for-sale, visibility, reports, category/item-wise permissions, enquiry ko Student mein convert karna	Principal / Owner
Permitted Teacher	Specific Library/Inventory permissions + jo items/categories direct-issue enabled hain; baaki sab par approval	Store-wali teacher (Inventory), library-wali teacher (Library)
Baaki Teachers	View-only — apni class ki items + apna "My Items"	Class ki craft supplies check karti teacher
Parent (enrolled)	Apne bacche ki issue/borrow/purchase items, return dates, wishlist	Borrowed book track karta parent
Public visitor	Sirf public-page items dekh sakta hai; kuch lene ke liye enquiry submit karta hai, jo Student conversion tak le jaati hai	Website dekh raha naya prospective parent

08 Key benefits

- Har item ke liye ek hi system — stationery ho ya library book — kuch untracked nahi rehta.
- Library alag system nahi, isi inventory ka hissa — ek panel, ek Admin control, ek report.
- By-default approval-required issue policy, jise Admin category ya single-item level par selectively relax kar sakta hai.
- Deposit-vs-price safety check se koi mehenga item galti se kam deposit par bahar nahi jaata.
- Available for Sale by-default OFF — koi item galti se bikne ke liye khula nahi rehta.
- Class-wise ya poori-category-wise free/chargeable setting se fee-structure flexibility.
- Public page visibility + wishlist se demand ka clear signal milta hai.
- Bulk upload ghanto ki entry bachata hai, safe review step ke saath.
- **Library aur Inventory ke liye alag-alag, category/item-wise permissions.**
- Bahar ke logon ke liye koi naya public system nahi — existing Enquiry process seedha Student conversion tak le jaata hai.
- Poora system ek hi baar mein, ek complete scope ke roop mein taiyar hota hai.

09 Conclusion

Ye module school ki har physical item ko — ek pencil se lekar library ki poori book collection tak — ek hi simple, transparent system ke andar laata hai. Library ko alag module banane ke bajaye, isi inventory ke andar ek specialized lending workflow ke roop mein rakha gaya hai, aur bahar ke logon ke liye access bhi existing, Admin-controlled Enquiry process se — seedha Student conversion tak — milta hai.

By-default approval-required issue policy, deposit-vs-price safety check, "Available for Sale" ka default-off rehna, class/category-wise free settings, public page visibility, aur wishlist jaise features isse aur safe aur flexible banate hain, jabki Library/Inventory ke liye alag category ya item-wise permissions Admin ko bilkul specific control dete hain.

Draft specification — figures, category names aur field names illustrative hain, development shuru hone se pehle final discussion mein confirm honge. (v4.0)